

NODE 5

Upper-Body Context Device (Chest-Mounted, Ring, or Wrist-Worn) Vibration, Haptic, and Somatic Feedback Integration

Field

This disclosure relates to wearable physiological context devices and, more particularly, to the integration of vibration, haptic, and somatic feedback systems within an upper-body context wearable configured to capture cardiovascular, autonomic, respiratory, thermal, and motion context for multi-node physiological interpretation.

Overview

In certain embodiments, the upper-body context device further incorporates one or more vibration or haptic actuation elements. These elements may be used to provide discreet somatic feedback, biofeedback, guided regulation cues, silent notifications, or cross-node signaling coordinated with other physiological monitoring devices.

The device may be configured as a chest-mounted wearable, a ring-form wearable, or a wrist-worn wearable, and may operate continuously during daytime, nighttime, exercise, stress, or sleep contexts.

Actuator Technologies

The vibration or haptic subsystem may include one or more of the following actuator types, individually or in combination:

- Low-profile Linear Resonant Actuators (LRA)
- Miniature Eccentric Rotating Mass (ERM) vibration motors
- Electromagnetic or voice-coil haptic actuators
- Piezoelectric haptic actuators
- Multi-zone micro-haptic arrays for spatially differentiated feedback

Actuators may be selected based on power efficiency, low audible output, and suitability for continuous wear.

Actuator Placement and Integration

One or more actuators may be integrated into the wearable housing, including but not limited to:

- Skin-contact surfaces to improve perceptibility at low amplitude
- Zoned placements enabling directional or patterned feedback
- Mechanically isolated compartments to reduce interference with ECG, PPG, or inertial sensors
- Strap-based or band-based integration points for improved tactile coupling

Functional Roles

The vibration or haptic subsystem may serve one or more functional purposes, including:

- Biofeedback correlated with heart rate, HRV, respiration, electrodermal activity, or thermal context

- Guided autonomic regulation cues (e.g., paced breathing or relaxation prompting)
- Silent alerts, event markers, or state change notifications
- Wear detection confirmation or contact integrity signaling
- Behavioral reinforcement or adherence prompting during extended monitoring

Cross-Node Coordination

In certain embodiments, the upper-body context device exchanges information with one or more additional physiological monitoring devices and provides haptic output based on multi-node state. Examples include:

- Haptic signaling synchronized with detected physiological events in other nodes
- Context-aware inhibition or modulation of actuation based on stress, sleep, motion, or elevated heart rate
- Coordinated feedback routines to support system-level interpretation and user comprehension

Closed-Loop Operation

In certain embodiments, actuation parameters are controlled in a closed-loop manner using real-time sensor input including ECG electrodes, PPG sensors, respiratory dynamics sensing, temperature sensors, electrodermal activity sensors, and inertial measurements. Closed-loop control may modulate vibration frequency, intensity, or pattern based on detected physiological state or user context.

Safety and Control

The device may include safety mechanisms such as maximum intensity limits, duty-cycle constraints, automatic shutoff conditions, user override controls, and context-aware inhibition to ensure safe, comfortable operation during continuous wear.